Threat Detection and Incident Response

Visual: Icon or graphic representing a monitoring system (e.g., a magnifying glass over a computer screen).

Attendance

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What is Threat Detection?

- **Definition:** Identifying signs of malicious activity or vulnerabilities within a network or system.
- **Key Techniques:**
  - Monitoring network traffic.
  - Using intrusion detection systems (IDS).
  - Log analysis.

    Visual: Diagram showing components of a detection system (e.g., logs, alerts, analysis).

What is Incident Response?

- **Definition:** The process of addressing and managing the aftermath of a security breach or attack.
- **Phases of Incident Response:**
  - Preparation
  - Identification
  - Containment
  - Eradication
  - Recovery
  - Lessons Learned

Tools and Strategies

- **Popular Tools:**
  - SIEM (e.g., Splunk, QRadar).
  - Intrusion detection systems (Snort, Suricata).
- **Strategies for Detection:**
  - Set up alerts for unusual behavior.
  - Regularly audit logs and configurations.
- **Proactive Measures:**
  - Conduct mock incident response drills.

    Visual: Screenshots or icons of key tools (e.g., Splunk dashboard).

Hands-On Activity Overview

Platform: TryHackMe

- **Activity:** "Threat Detection" room or equivalent module.
- **Concepts Covered:**
  1. Analyzing network traffic for anomalies.
  2. Identifying signs of compromise.
- **Steps:**
  1. Complete initial tasks as a group.
  2. Spend 15-20 minutes working independently.
  3. Discuss findings as a class.

     Visual: Screenshot of the TryHackMe activity interface.

Interactive Activity: Simulated Incident Response

- **Objective:** Work in teams to handle a simulated cybersecurity incident.
- **Setup:**
  - Scenario: A mock attack on a company network (e.g., data exfiltration detected).
  - Provide logs and alerts for analysis.
- **Roles:**
  - Incident Responders: Analyze logs and propose immediate actions.
  - Observers: Take notes and provide feedback during the debrief.
- **Steps:**
  - Identify the breach and affected systems.
  - Propose containment and eradication steps.
  - Share findings with the class.

  Visual: Simplified network map with simulated attack details.

Debrief and Key Takeaways

- **Discussion Points:**
  - What were the key signs of the incident?
  - Which detection strategies worked best?
  - How would you improve your response next time?
- **Key Takeaways:**
  - Early detection is critical to mitigating damage.
  - Incident response is a collaborative effort.

  Visual: Bullet-point summary or checklist.

Q&A Session

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Join us for the next sessions

Thursdays: 11AM – 12PM & 2PM – 3PM
Friday: 3PM – 4PM
Monday: 12PM – 1PM

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